MAX D. SCHNEIDER

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PROFESSIONAL SUMMARY AND SKILLS

Applied statistician brings 15+ years of expertise leading teams to build right-sized statistical and machine learning models for operational systems, particularly for natural hazards. Expert in quantifying uncertainty, evaluating model performance, and visualizing risk under uncertainty using real and synthetic data. Frames and solves ambiguous technical problems to support users to characterize and reduce risks. Proven leadership of multidisciplinary teams to drive science-based decision-making across engineering, research, and client-focused domains.

- ***Languages & tools***: R, Python, C++, bash, MySQL, Git, HPC/Supercomputing, GIS, Shiny, ggplot2
- ***Models & methods***: Bayesian modeling, machine learning, linear/multilevel regression, time series models, spatial/spatiotemporal models, clustering, boosted regression, model validation, A/B testing
- ***Technical communication***: user engagement, needs assessment, usability studies, data visualization/mapping
- ***Team leadership***: project vision, management and execution across disciplines, skillsets and countries (proficient in Spanish, German and Russian)

EDUCATION

University of Washington, Seattle *September 2016 – December 2021*
Doctor of Philosophy in Statistics

University of Potsdam (Potsdam, Germany) *October 2012 – February 2016*
Master of Science in Mathematics: Mathematical Modeling and Data Analysis

University of California, Los Angeles *September 2007 – June 2011*
Bachelor of Science in Statistics and Human Complex Systems

PUBLICATIONS

1. **Schneider, M.**, Wein, A., McBride, S.K., van der Elst, N., Becker, J., Castro, R., Diaz, M., Gonzalez-Huizar, H., Hardebeck, J., Michael, A., Mixco, L. and Page, M. Meet the people where they are: Assessing user needs for aftershock forecast products in El Salvador, Mexico and the United States, *Int. J. of Dis. Risk Reduc.*, 125, 2025.
2. **Schneider, M.**, Barall, M., Guttorp, P., Hardebeck, J., Michael, A., Page, M. and van der Elst, N. Bayesian ETAS modelling for the Pacific Northwest: Uncovering effects of tectonic regimes, regional differences and swarms on aftershock parameters, *Bull. of the Seism. Soc. of America*, 115, 2219–2236, 2025.
3. **Schneider, M.** and Guttorp, P. What do we know without the catalog? Eliciting prior beliefs from experts for aftershock models, *The Seis. Record*, 4, 259–267, 2024.
4. Mizrahi, L., Dallo, I., ..., **Schneider, M.**, Page, M., ..., Wiemer, S. Developing, testing, and communicating earthquake forecasts: Current practices and future directions, *Rev. of Geophys.*, 62, 2024.
5. Heinrich-Mertsching, C., Thorarinsdottir, T., Guttorp, P. and **Schneider, M.** Validation of point process predictions with proper scoring rules, *Scandinavian J. of Stat.*, 51, 2024.
6. **Schneider, M.**, Flury, H., Guttorp, P. and Wright, A. Earthquake catalog processing and swarm identification for the Pacific Northwest, *Seis. Res. Lett.*, 94, 2500–2513, 2023.

7. **Schneider, M.**, Wein, A., van der Elst, N., McBride, S.K., Becker, J., Castro, R., Diaz, M., Gonzalez-Huizar, H., Hardebeck, J., Michael, A., Mixco, L., Page, M. and Palomo, J. Visual communication of aftershock forecasts based on user needs: A case study of the United States, Mexico and El Salvador, *IEEE Vis. for Communication*, 1, 2023.

8. **Schneider, M.**, Cotton, F. and Schweizer, P.-J. Criteria-based visualization design for hazard maps, *Nat. Haz. and Earth Sys. Sci.*, 23, 2505–2521, 2023.

9. **Schneider, M.**, McDowell, M., Guttorp, P., Steel, E.A. and Fleischhut, N. Effective uncertainty visualization for aftershock forecast maps, *Nat. Haz. and Earth Sys. Sci.*, 22, 1499–1518, 2022.

10. Levers, C., **Schneider, M.**, Prishchepov, A., Estel, S. and Kümmerle, T. Spatial variation in determinants of agricultural land abandonment in Europe. *Sci. of the Total Env.*, 644, 95-111, 2018.

11. Strader, A., **Schneider, M.** and Schorlemmer, D. Prospective and retrospective evaluation of five-year earthquake forecast models for California. *Geophys. J. Int.*, 211, 239-251, 2017.

12. **Schneider, M.**, Clements, R., Rhoades, D. and Schorlemmer, D. Likelihood- and residuals-based evaluation of medium-term earthquake forecast models for California, *Geophys. J. Int.*, 198, 1307-1318, 2014.

Manuscripts in preparation/review:

13. **Schneider, M.**, Wein, A., McBride, S.K., Artigas, B., Castro, R., Gonzalez-Huizar, H., Page, M. and van der Elst, N. From user needs to design choices: A method to develop user-centered forecast products. Manuscript in review.

14. **Schneider, M.**, Barall, M., Hardebeck, J., Michael, A., Page, M. and van der Elst, N. What makes a good aftershock forecast? Evaluating the components of the US Geological Survey aftershock forecasting system. Manuscript in preparation.

EXPERIENCE

Independent Statistical Consultant May 2026 –
Worldwide

- Institute of Geophysics and Volcanology, Rome, Italy. Develop operational forecasting and reporting systems based on emergency management needs for national Italian earthquake forecasts
- California Geological Survey, Sacramento, CA. Design human-centered color palettes and legends that can reach key messaging targets for California's state seismic hazard map

US Geological Survey May 2022 –
Research Statistician *Moffett Field, CA*

- Identify improvements to USGS aftershock forecasting operations by testing forecasts against observed aftershocks, developing statistical metrics to quantify distinct attributes of forecast skill
- Use high-performance computing to efficiently calculate and test millions of probabilistic forecasts following earthquakes across California and other tectonic regions
- Develop and test actionable maps, GIS systems and interactive software for aftershock forecasts, based on user needs elicited from stakeholders in engineering, utility and risk management sectors (among others) to support post-earthquake recovery and response decisions
- Mentor interns and lead 20+ member research team with collaborators across USGS's Earthquake Hazards Program and partner agencies in Mexico, El Salvador and New Zealand
- Author of four funding proposals, receiving $500K award from Bureau for Humanitarian Assistance

German Research Center for Geosciences August 2020 – May 2022
Statistical Consultant with Dr. Fabrice Cotton *Potsdam, Germany*

- Developed criteria for classifying seismological data to improve reproducibility of natural hazard maps
- Created method for choosing color palettes for human-centered natural hazard mapping
- Identified key targets when communicating seismic hazards to engineers to support science-based building codes and seismic resilience
- Redesigned seismic hazard map for Germany based on above methods and evaluated its efficacy in communicating key targets in a controlled task-based experiment

UW Department of Statistics June 2016 – December 2021
Graduate Research Assistant with Dr. Peter Guttorp *Seattle, WA*

- Implemented research-grade pipeline of probabilistic earthquake forecasting and communication: from earthquake data processing to modeling and uncertainty quantification, through to forecast and uncertainty visualization
- Built Bayesian spatiotemporal point-process models of earthquake rates, including statistical elicitation of priors from experts and quantification of multiple sources of uncertainty
- Applied models to the Pacific Northwest to infer how regional characteristics affected model parameters, using scripts written in R, Python and C++
- Created automated algorithms to classify earthquake clusters between aftershock sequences and swarms to improve earthquake rate modelling of complex tectonic environments
- Evaluated techniques for uncertainty visualizations for aftershock forecast maps, using controlled experiments to study how visualization techniques affect users' performance in real-world decision tasks

UW Center for Statistics and the Social Sciences September 2019 – June 2020
Statistical Consultant *Seattle, WA*

- Delivered statistical and analytical consulting emphasizing defensibility, documentation, and reproducibility of modelling and other data analyses
- Advised dozens of clients on experimental design, regression modeling, machine learning/artificial intelligence, time-series analysis, uncertainty quantification, spatial/spatiotemporal modeling, engineering statistics, wave analysis and numerical models
- Guided researchers in best practices for right-sized data analyses, focusing on technical summaries and interpretations to improve their communication to non-specialists

NOAA National Centers for Environmental Prediction June 2019 – August 2019
Graduate Intern *College Park, MD*

- Quantified and visualized prediction errors for NOAA ensemble ocean wave model, supporting hazard modelling for coastal and oceanic environments
- Used Python to build time series models of wave height prediction errors during Hurricane Irma (2017)
- Advised on error quantification and propagation for coupled operational modelling systems to NOAA scientists and interagency coastal hazards group

UW Department of Statistics and CSSS September 2017 – March 2019
Teaching Assistant with Drs. June Morita and Elena Erosheva *Seattle, WA*

- Taught graduate coursework in statistical methods for longitudinal data and undergraduate coursework in mathematical statistics, instructing weekly labs/problem sessions
- Conceived, prepared solutions for and graded weekly homework assignments, quizzes and exams

Humboldt University of Berlin, Geography Institute April 2015 – September 2016
Research Assistant with Drs. Tobias Kümmerle and Tobia Lakes *Berlin, Germany*

- Built machine learning-based classifiers of spatial environmental processes across large datasets to inform evidence-based decision-making
- Reviewed statistical literature on the boosting algorithm and proved mathematical optimality of specific choices for the algorithm's three components
- Investigated spatial relationships between predictor and response variables of land use changes in Europe using spatially-varying models

Karlsruhe Institute of Technology, Geophysical Institute August 2013 – December 2013
Research Assistant with Dr. Bijan Khazai *Karlsruhe, Germany*

- Analyzed direct and indirect economic losses from past earthquakes in Japan, investigating sector-specific losses for hazard and risk assessments for engineering and policy applications
- Provided statistical consulting to group of earthquake engineers and economists

Human Rights Data Analysis Group August 2010 – December 2011
Statistics Intern *Palo Alto, CA*

- Performed algorithmic data cleaning and processing for sample of documents from Guatemalan National Police Archives, to be analyzed for court trials regarding past human rights violations
- Conducted field visit to National Police Archives to prepare and analyze data for statistical evidence being used in court cases

UCLA Department of Atmospheric and Oceanic Sciences March 2008 – January 2011
Research Assistant with Dr. Jochen Stutz *Los Angeles, CA*

- Conducted fieldwork and computational modelling to determine whether a spectrometry instrument could sufficiently measure the concentration of ozone and other compounds in the stratosphere

SELECTED AWARDS

- US Geological Survey STAR Awards (February 2023, March 2023, February 2024)
- USGS-Bureau of Humanitarian Assistance Authorization Request (post-doctoral research grant "Development and Deployment of Operational Aftershock Forecasts for Situational Awareness during Earthquake Crises") ($491,606, 2022-2026)
- Seismological Society of America Annual Meeting Student Presentation Award (April 2021)
- USGS Earthquake Hazards Program External Grant ($99,863, 2020-2021)
- University of Washington Statistics Department Fellowship, to pursue PhD at UW (September 2016)
- Humboldt University Research Track Scholarship (€5000), for pre-doctoral research (Feb.–Aug. 2016)

ACADEMIC MEMBERSHIP, SERVICE AND ADDITIONAL INFORMATION

Professional service: Associate editor of Journal of Science Policy and Governance; chair of the US Geological Survey Social Science Working Group; reviewer for Bulletin of the Seismological Society of America, Seismological Research Letters, International Journal of Disaster Risk Reduction and others; member of the Seismological Society of America Meetings and Ethics Committees

Leadership experience: UW Statistics in the Community Moderator (pro bono statistical consulting group, 2019-2021); UW Space-Time Reading Group, Co-Organizer (2018-2020)

Professional membership: American Statistical Association, Seismological Society of America

Multilingual: Native speaker of Russian and English, proficient in Spanish and German

REFERENCES

Ruth Harris, Research Geophysicist at the US Geological Survey.
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Fabrice Cotton, Section Head of Seismic Hazard and Risk Dynamics at the German Research Center for Geosciences.
Telegrafenberg, 14473 Potsdam, Germany
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Andrew J. Michael, Research Geophysicist at the US Geological Survey.
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ajmichael@usgs.gov, 650-439-2777